STATISTICS · ANTICIPATING PATTERNS

Conditional Probability: Screening Tests

Explore & Reflect

Name _____ Date _____ Period _____

Open **Conditional Probability: Screening Tests** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A test is 99% sensitive and 99% specific, and 1% of people have the condition. If you test positive, guess the chance that you have it.

2. Is P(positive | condition) the same as P(condition | positive)? Explain.

2 Explore use the tool

1. Set prevalence 1%, sensitivity 99%, specificity 99%. Record the true positives, false positives and P(C | +).

2. Keep the 99%/99% test. Which prevalence on the slider is the first to make P(C | +) at least 90%?

3. Prevalence 5%: compare raising sensitivity to 100% with raising specificity to 99.5%. Which helps P(C | +) more? Why?

4. Set sensitivity 50% and specificity 50%. What is P(C | +)? Why?

3 Record use the tool and fill in the table

P(condition | positive) and P(condition | negative)

Prevalence	Sensitivity	Specificity	P(C | +)	P(C | −)
0.1%	99%	99%		
2%	90%	95%		
20%	90%	95%		
50%	80%	60%		

4 Explain what did you discover?

1. P(C | +) = _____ ÷ _____.
2. When a condition is rare, many positive results are _____.
3. Events A and B are independent when P(A | B) = _____.
4. Explain to a patient why a positive result on a 99%-accurate test (99% sensitive and specific) for a disease that affects 0.1% of people is probably a false positive. Use counts out of 100,000.

5 Apply a new situation, no tool

1. At a school, 45 of 60 seniors and 30 of 90 juniors drive to school. Find P(drives | senior) and P(senior | drives). Are "senior" and "drives" independent?
2. P(A) = 0.3, P(B | A) = 0.4 and P(B | not A) = 0.2. Find P(A | B).
3. 40% of email is spam. A filter flags 95% of spam and 2% of other email. Find P(spam | flagged).